QUAKELOGIC

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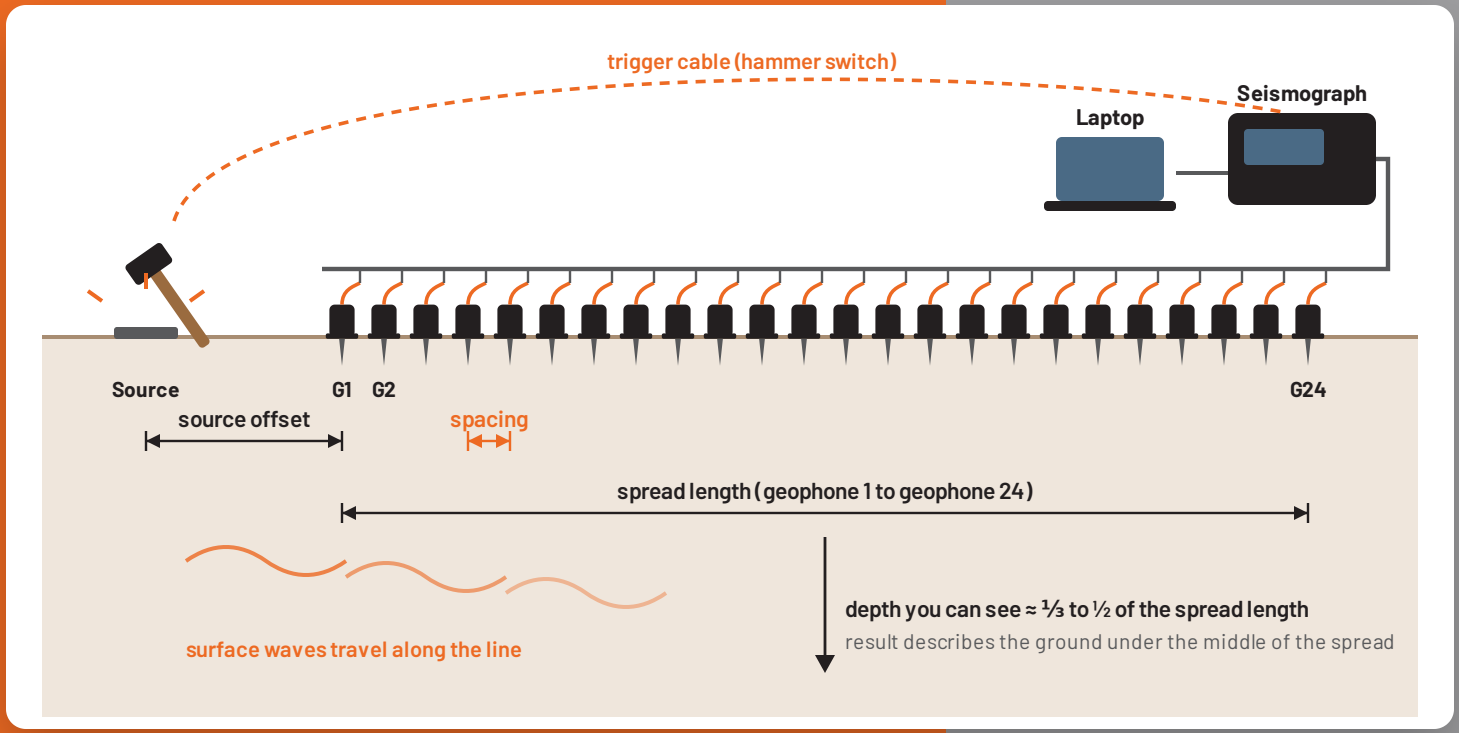
MASW FIELD GUIDE

MASW STUDIO

SET UP · RECORD · ANALYSE

Equipment, line layout, seismograph settings and MASW Studio, step by step, for your first survey.

MASW STUDIO 1.3.2 · WINDOWS 11



FIELD GUIDE

From the equipment case to a Vs report

FOR FIRST-TIME MASW USERS · MASW STUDIO 1.3.2

1 Before you start
2 Plan the line
3 Lay out the spread
4 Connect the seismograph and the trigger
5 Seismograph settings and recording
6 Sloping ground, riverbeds and valleys
7 Before you leave the site
8 Install MASW Studio and analyse the data
9 Field checklists and field sheet
10 Troubleshooting

1. Before you start

This guide takes you from an empty equipment case to a finished shear-wave velocity (Vs) report. It assumes no previous experience. Read it once in the office, then take the checklist pages (section 9) to the field.

What MASW measures

When a hammer hits the ground, it sends out surface waves that roll along the ground like ripples on water. Their low frequencies (long waves) reach deep into the ground, and their high frequencies (short waves) stay shallow. Each frequency therefore travels at the speed of the ground it passes through.

A line of geophones records these waves. MASW Studio measures the speed of each frequency and turns those speeds into a profile of shear-wave velocity (Vs) against depth. Vs tells an engineer how stiff the ground is. It is used for seismic site class (Vs30), foundation design, compaction checks and finding soft zones or voids.

> **You need three things to be right in the field:** a straight line of well-planted geophones at a measured spacing, a clean trigger at the moment of each hit, and the positions written down. Everything else can be fixed in the software afterwards.

The equipment list

ITEM	WHAT TO BRING	NOTES
Seismograph	Engineering seismograph with 12, 24 or 48 channels	24 channels is the usual MASW setup
Laptop	Laptop with the seismograph's own recording software	Charged, with a spare battery or power bank
Geophones	One per channel, **4.5 Hz vertical** geophones with spikes	4.5 Hz reach much deeper than 10 Hz or 14 Hz geophones
Geophone cable	Spread cable (takeout cable) with a takeout every few metres	The takeout spacing sets the largest geophone spacing you can use
Trigger	Hammer switch (trigger switch taped to the handle) or trigger geophone, with its cable	Starts the recording at the instant of the hit
Source	Sledgehammer, **5–10 kg (10–20 lb)**	Heavier hammers reach deeper (see section 2)
Strike plate	Aluminium or hard plastic plate, about 20–25 cm (8–10 in) across, 2–3 cm thick	Gives a sharp, repeatable hit
Measuring tape	50 m or 100 m (150 ft or 300 ft) tape	Plus two pegs to hold it straight
Markers	Survey flags, spray paint or pegs	Mark each geophone position
Power	Charged 12 V battery or the seismograph's power pack, with cables	Check charge the day before
Elevation (sloping sites)	Level and staff, or GNSS/GPS receiver	Only needed when the ground is not level (section 6)
Field sheet	Printed field sheet (section 9) and a pencil	Write down every file
Safety	Ear defenders, safety glasses, gloves, safety boots, high-visibility vest	The hammer operator needs all of them
Spares	Spare spikes, a small hammer to plant spikes, tape, cable ties	Small things that end a day early

Check everything in the office

1. Charge the seismograph battery, the laptop and any GNSS receiver.
2. Connect the seismograph to the laptop and start the recording software once, so you know it works before you are on site.
3. Connect the trigger and tap the hammer on the floor: the software must start a recording on every tap.
4. Install MASW Studio on the computer you will use for analysis (section 8). It does not need to be the field laptop.

2. Plan the line

Pick the place

- Choose a **straight line** across the area you want to know about. The result describes the ground under the middle of the geophone line.
- Prefer **level, open ground**. A gentle, even slope is fine (section 6). Avoid lines that cross a ditch, a wall, a kerb or a change from soil to pavement.
- Keep away from **noise**: running engines, generators, busy roads, pumps, trees moving in strong wind and people walking near the geophones. If you cannot avoid traffic, record between vehicles.

- Keep at least a few metres from **buried pipes, basements and foundations** if they are not what you are looking for.

How deep can you see?

The depth you can see depends mainly on the **length of the geophone line** (the spread) and on how much low-frequency energy the source produces. A good rule of thumb: you can usually see to about **one-third to one-half of the spread length**.

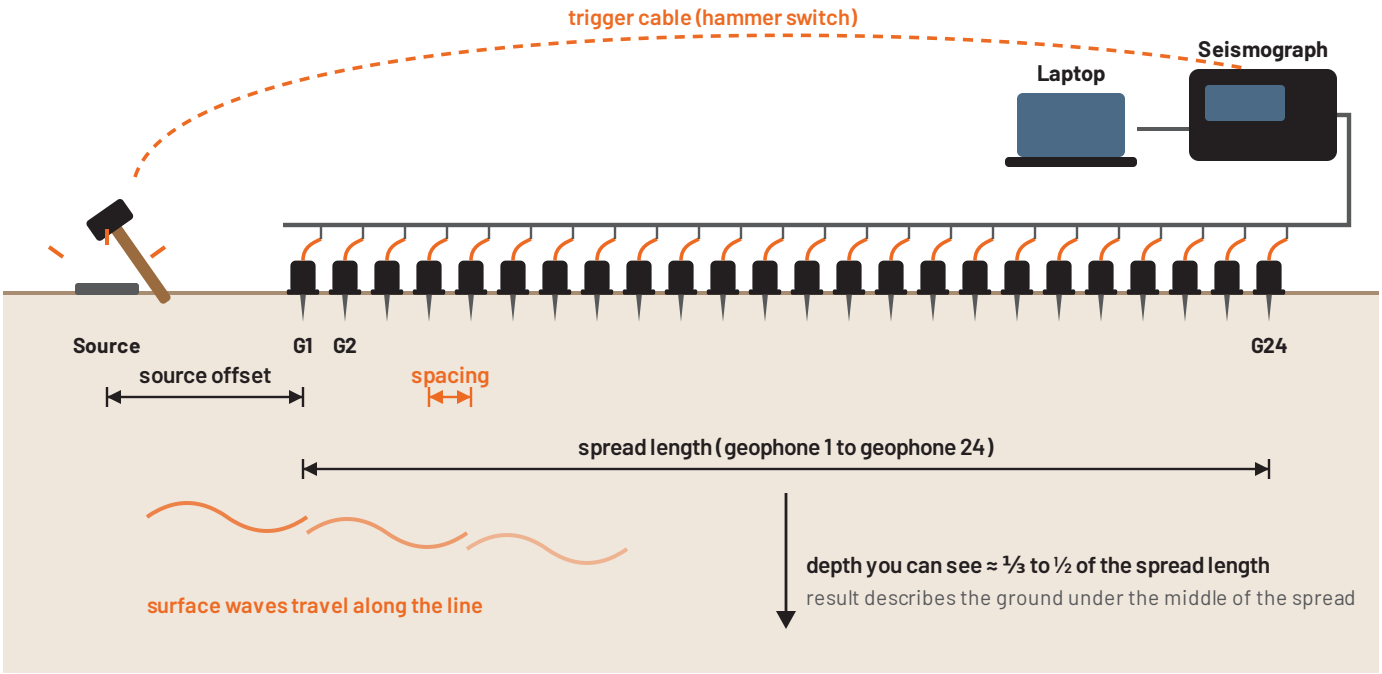
WANTED DEPTH	GEOPHONE SPACING (24 CHANNELS)	SPREAD LENGTH	SOURCE
5–10 m (15–30 ft)	1 m (3 ft)	23 m (75 ft)	5 kg (10 lb) hammer
10–20 m (30–65 ft)	2 m (6.5 ft)	46 m (150 ft)	5–10 kg (10–20 lb) hammer
20–30 m (65–100 ft), e.g. Vs30	3 m (10 ft)	69 m (225 ft)	10 kg (20 lb) hammer, or a weight drop, plus stacking

- **Closer spacing** shows more detail near the surface.
- **A longer spread** sees deeper, but the far geophones get weaker signals.
- Vs30 needs the model to reach 30 m. On soft ground that usually needs the longest spread in the table and good stacking.

Where the hammer goes

The hammer point (the **source**) is on the same line, a short distance before the first geophone. This distance is the **source offset**.

- Use about **one-quarter of the spread length** (5–10 m for a 46 m spread).
- Too close (less than one spacing) and the first geophones are swamped by the hit. Too far and the far geophones hear very little.
- Recording a second shot from the **other end** (the reverse shot) is good practice: it costs five minutes and gives a second, independent profile.



SIDE VIEW OF A MASW LINE: HAMMER POINT, GEOPHONES, CABLES AND THE SEISMOGRAPH

3. Lay out the spread

1. **Mark the line.** Stretch the tape along the line and fix both ends. Decide which end is **geophone 1** (normally the end nearest the hammer point) and write it on the field sheet.

2. **Mark the positions.** Put a flag or paint mark at every geophone position (for example 0, 2, 4 … 46 m) and at the hammer point.
3. **Lay the cable.** Lay the spread cable along the line with takeout 1 at geophone 1. Keep it flat on the ground; a cable flapping in the wind adds noise.
4. **Plant the geophones.** At each mark, scrape away grass and loose topsoil, then push the geophone in by hand until the base sits on the ground. Tap the spike in with a small hammer if needed, but never hit the geophone case.
5. **Check they are upright.** A vertical geophone must stand straight (within about 10–20° of vertical, depending on the geophone type; check its data sheet). A tilted or loose geophone gives a weak, noisy trace.
6. **Connect in order.** Clip geophone 1 to takeout 1, geophone 2 to takeout 2, and so on. A swapped pair shows as two traces out of order in the gather.



Upright, spike fully in, base on firm soil

Tilted: weak, distorted trace

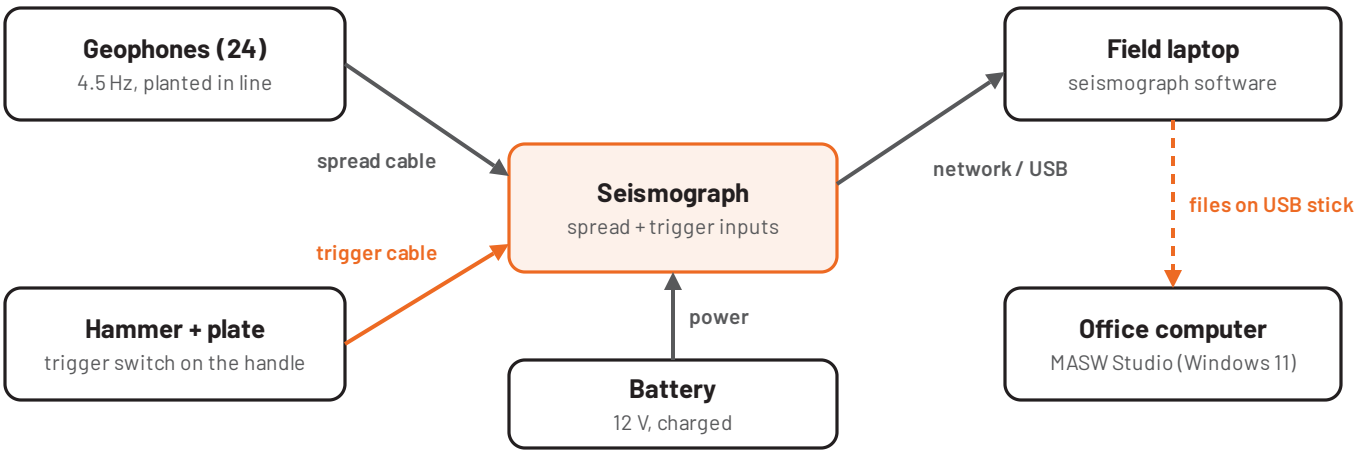
Not pushed in, or on grass and loose topsoil

PLANT EVERY GEOPHONE UPRIGHT AND FIRM; TILTED OR LOOSE GEOPHONES SPOIL THE RECORD

On pavement or concrete you cannot use spikes. Use geophone base plates or tripod stands, or sit the geophones on a small bag of sand. Every geophone must still be firmly in contact with the ground.

4. Connect the seismograph and the trigger

1. Connect the **spread cable** to the seismograph.
2. Connect the **trigger cable** to the seismograph's trigger input. For a hammer switch, tape the switch firmly to the hammer handle, just below the head. For a trigger geophone, plant it next to the strike plate.
3. Connect the **battery** and switch the seismograph on.
4. Connect the **laptop** (network cable or USB, depending on the seismograph) and start its recording software.
5. **Test the trigger.** Hit the plate lightly. The software must start one recording per hit, and the first geophone should show the arrival right at the start of the record. If nothing happens, check the trigger cable and the trigger sensitivity setting.
6. **Test every channel.** Tap the ground lightly next to each geophone, or look at the noise display if the software has one. Every channel must show signal. A flat (dead) trace usually means a loose clip or a damaged geophone.



WHAT CONNECTS TO WHAT

5. Seismograph settings and recording

Settings to use

Enter these in the seismograph's recording software before the first shot:

SETTING	USE	WHY
Sample interval	0.5 ms or 1 ms (2,000 or 1,000 samples per second)	Plenty for surface waves (they are below about 100 Hz)
Record length	1 s; 2 s for long spreads or very soft ground	The slow surface waves must reach the last geophone
Pre-trigger / delay	0	Time zero is the moment of the hit
Stacking	5–10 hits per shot	Adds the hits together; random noise cancels out
Acquisition filters (low-cut, notch)	Off	Filtering, if needed, is done later in MASW Studio
Geometry	Geophone spacing, position of geophone 1 and the source position	Saved in the file; MASW Studio reads it on import
File format	SEG-2 (usually `.sg2`, `.seg2` or `.dat`)	Read directly; SEG-Y and DRM (`.drm`) also work

Record a shot

1. Make sure nobody is walking near the line and no vehicle is passing.
2. Place the strike plate at the hammer point. The hammer operator wears ear defenders, safety glasses and gloves; everyone else stands back.
3. Hit the centre of the plate squarely, with a full swing. Keep hits the same from one to the next.
4. **Look at the record** after each hit (see below). Repeat any hit spoiled by noise; most software lets you undo the last stack.
5. When the stack is complete, **save the file** and write its number, the source position and any notes on the field sheet.
6. Move the plate to the other end for the reverse shot, if planned, and repeat.

> **Want a 2D section?** Either move the whole spread along the line by a few geophone spacings after each shot (a roll-along) and record again, or record one long spread with many channels. MASW Studio builds a 2D section from either.

What a good record looks like

- **Good:** every trace shows a clear, slanting band of large, slow wiggles (the surface waves), arriving later at each farther geophone.
- **Dead channel:** one flat trace. Check the clip and the geophone.
- **Noisy channel:** one trace full of fast jitter. Re-plant that geophone and check for wind or someone standing nearby.
- **Clipped:** waveforms with flat tops on the near geophones. Hit a little more softly, or reduce the seismograph's gain for the nearest channels.
- **Weak far traces:** the far geophones show almost nothing. Stack more hits, hit harder, or use a heavier source.

6. Sloping ground, riverbeds and valleys

Choose level ground when you can. When the geophones cannot all be at the same height (on a slope, across a riverbed or a valley), the waves travel along the ground, which is longer than the flat distance. On a 30° slope the flat distance is 13 % shorter than the real path.

1. Lay the tape **along the ground** and mark positions along the tape as usual.
2. Measure the **elevation of every geophone and of the hammer point** with a level and staff, or a GNSS/GPS receiver. Write them on the field sheet.
3. If the seismograph software has an elevation (z) field for each channel, enter them there too, so they are saved in the file.

> **Coming in MASW Studio 1.3.3:** elevations are read from SEG-2, SEG-Y and DRM files, or entered under **Dataset → Ground elevation**. Distances are then measured along the ground, and the 2D section follows the shape of the ground. Until then, choose the most level line you can.

7. Before you leave the site

- Copy every file to a USB stick **and** keep it on the laptop: two copies.
- Check the field sheet: every file has a source position, spacing, the end of geophone 1 and notes.
- Photograph the line from both ends with the geophones in place. Photos can go in the MASW Studio report.
- Note the GNSS/GPS coordinates of both ends of the line.
- Pick up every geophone, flag and peg. Count the geophones before you leave.

8. Install MASW Studio and analyse the data

Install

1. Download **MASW Studio Setup** from the QuakeLogic product page, on a computer running Windows 11 (64-bit).
2. Run the setup file. Windows may show **Unknown publisher**: choose **More info → Run anyway**.
3. Follow the installer and leave **Launch MASW Studio** selected on the last page. The full version opens; there is no licence key and no internet connection is needed.
4. Next time, start it from **Start → QuakeLogic → MASW Studio**.

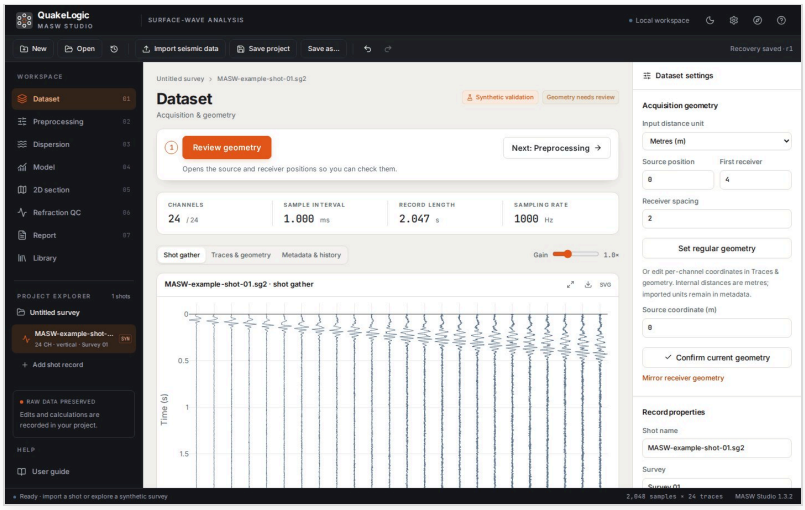
On first launch the **help wizard** offers a tour. Take it once; it shows every step below on a built-in example recording.

Analyse your recording

The orange button in the bar at the top of each page is always the next thing to press.

1. **Import.** Choose **Import seismic data** and select your file (or several files). The import summary shows the channels, the sampling and the positions read from the file.

2. **Dataset: check the geometry.** Compare the positions with your field sheet. If they are wrong or missing, enter the source position, first geophone position and spacing under **Set regular geometry**. Then choose **Confirm geometry**.
3. **Preprocessing (optional).** A 5–80 Hz band-pass filter is preset. Preview it, and apply it if the record looks cleaner.
4. **Dispersion.** Choose **Calculate dispersion**. Click along the **bright band** from low to high frequency; six to ten points are enough. The band normally falls from left to right.
5. **Model.** Run the inversion, check that the model curve follows your points, then accept it. Enter the interpretation depth (30 m for Vs30).
6. **2D section.** With several shots along a line, or one long record, build the section.
7. **Report.** Enter client, site and operator, add the site photos, preview the PDF and save it. Choose **Save project** to keep the analysis.



DATASET PAGE: THE ORANGE BUTTON IN THE STEP BAR IS THE NEXT ACTION

The full user guide (in the software under **User guide**, and as a PDF on the product page) explains every setting.

9. Field checklists and field sheet

Before you leave the office

- Seismograph, laptop, battery and chargers, all charged
- Geophones (one per channel, plus two spares) and the spread cable
- Trigger switch or trigger geophone, and the trigger cable
- Sledgehammer, strike plate, tape, flags, pegs, small hammer
- Level and staff or GNSS receiver (sloping sites)
- Printed field sheets, pencil, safety equipment
- Trigger tested in the office

On site, before the first shot

- Line straight, positions marked and written down
- Geophone 1 end decided and noted
- Every geophone upright, firm and connected in order
- Trigger works on every hit
- Every channel shows signal; no dead or noisy channels
- Settings: 0.5–1 ms sampling, 1–2 s record, no pre-trigger, filters off, geometry entered

Field sheet

FILE	LINE	SOURCE POSITION (M)	GEOPHONE 1 AT (M)	SPACING (M)	HITS STACKED	NOTES (NOISE, WEATHER, ELEVATIONS)

10. Troubleshooting

PROBLEM	LIKELY CAUSE	WHAT TO DO
Nothing records when the hammer hits	Trigger cable loose, or trigger sensitivity too low	Check the cable and switch; raise the trigger sensitivity
Recording starts at random	Trigger sensitivity too high, or trigger cable moving	Lower the sensitivity; tape the cable to the handle
One trace is flat	Loose clip or broken geophone	Re-clip; swap the geophone
One trace is very noisy	Geophone loose, tilted, or in grass or wind	Re-plant firmly and upright
Far traces are almost empty	Not enough energy	Stack more hits, hit harder, use a heavier source
The bright band fades out below 10–15 Hz	Not enough low-frequency energy, or 10/14 Hz geophones	Heavier source, more stacking, 4.5 Hz geophones, longer spread
MASW Studio says the geometry is missing	Positions were not entered in the seismograph	Enter them under Dataset → Set regular geometry
Picks follow a straight rising line	A dashed guide line was picked, not the band	Choose **Reset picks** and click along the bright band
A DRM file with pre-trigger is refused	The recording used a pre-trigger or channel delay	Record with pre-trigger 0, or export the file as SEG-2

Help: email support@quakelogic.net from the **Support** menu in MASW Studio. Email response time: 3 business days.

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MASW STUDIO

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Safety first

The hammer operator wears ear defenders, safety glasses, gloves and safety boots. Keep everyone clear of the swing, and never work near traffic without a lookout.

About QuakeLogic:

QuakeLogic Inc. is a California-based engineering and technology company providing innovative, reliable, high-quality solutions to customers worldwide.

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